

Description of China Household Finance Survey Data

Variable Rules

2011-2013

I. Data Cleaning

1. Cleaning process

After the data is collected, the center will conduct the preliminary processing on all data, mainly including: deleting invalid samples due to serious groundless answer and cheating by interviewers, deleting invalid variables, deleting sensitive data, correcting duplicate sample numbers caused by humans, and correcting interviewers' active reports that are man-made misuse; merging, tracking new visiting data, splitting household and personal data, splitting multiple choices; adding labels, adding questionnaire types; cleaning up notes and other options, confirming the numerical value interpolation and related rules.

After preliminary processing, the data generates a usable version. If the extreme value or abnormal value is found again during the data utilization, the center will check the secondary recording for confirmation. If there is no recording, the extreme value and abnormal value are processed by creating a model to update the data version and inform the users.

2. Data storage

After completing the data cleaning, the data of China Household Finance Survey can be stored in the following 3 data sets.

1. The data set file name contains “**hh**”, which represents the data of the household part within the questionnaire, such as: assets and liabilities, household expenditures and income, financial knowledge, grassroots governance and subjective evaluation;
2. The data set file name contains “**ind**”, which represents the data of the personal part within the

questionnaire, such as: demographic characteristics (partial), personal work and income information, insurance and security, family member education, etc.

3. The data set file name contains “**master**”, which represents the regional data, specifically including the province information of the investigation and the weight of each investigation data.

In addition, the original data usage format corresponds to the dta format (which can be opened with stata13.0 [best version] or above).

II. Variable Naming Rules

The variable name of the data of China Household Finance Survey questionnaire consists of the initial letter and the last four digits. The different initials correspond to different parts of the questionnaire (e.g. b-agriculture/business; d-financial assets), some special identification variables (ID variables, urban and rural areas, provinces, etc.) are renamed according to their meaning for use. Some variables are followed by the suffix letter it after the original variable name, indicating the inquiry of the range of values for the previous question. Each question in the questionnaire is given the corresponding variable name. The variables that need special attention and their naming rules are described as follows:

1. ID variables

ID variables contain hhid and pline.

The hhid identifies the household variable. Each hhid represents a household, and can exclusively identify a household. The last four letter of hhid represents the investigation year, for example, hhid_2011 represents the sample number visited in 2011. The pline is to identify the variables of family members in each family. Each pline represents a family member; hhid and pline can uniquely identify individuals, and the pline of the same family members of the same family remain the same at different times.

2. Weight (swgt)

Under our sampling design plan, the number of households represented by each household is different because the probability of each household being drawn is different. When inferring the total, it is necessary to adjust the weights to truly and accurately reflect the number of households represented by each household sample to obtain a correct inference of the total. All calculation results of the China Household Finance Survey have been adjusted regarding the sampling weights. There is a weight in the master data set: swgt.

The calculation method of the sampling weight is as follow: according to the sampling of each stage, the probability P_1 to be drawn of the city and the county under investigation, and the probability P_2 to be drawn of the community (village) under investigation in its district or county and the probability P_3 to be drawn of the sample under investigation in its community (village) are calculated respectively. Three-stage sampling weights are calculated respectively $W_1=1/P_1$, $W_2=1/P_2$, $W_3=1/P_3$, and finally the sampling weight of the sample is $W=W_1 \times W_2 \times W_3$.

That is swgt = the reciprocal of the probability that this community is drawn.

To arrive at a more accurate conclusion, we recommend that the data user applies the sampling weight "swgt" for analysis.

3. Other options in the variables

In data variable single-choice or multiple-choice questions, the last option in general is "Others (please specify it)". If the respondent chooses this option, there will be one more fill-in-the-blank question. In the 2011 and 2013 China Household Finance Survey data, the newly generated variable naming format is: the original variable followed by the suffix p, for example: a4002p indicates that the information that the respondent mainly focuses on is other related information.

4. Cyclic questions

For all cyclic questions, the naming convention is to add the suffix _# after the original variable name; # represents the # cyclic times. For example: c2003_1 indicates the floor area of the first set of house; thus c2003_2 indicates the floor area of the second set of house.

In addition, the questionnaire also contains other cycles generated by the multi-choice option loading, for instance, c5005_1 indicating c5002 option 1 agricultural land loading, c5005_2 indicating option 2 rural collective construction land loading.

5. Multiple-choice questions

For multiple-choice questions, the processing principle is to convert each option to a dummy variable with values 0 and 1. Multiple-choice questions are divided into two categories: acyclic multiple-choice questions and cyclic multiple-choice questions.

(1) Acyclic multiple-choice questions

The naming rule for acyclic multiple-choice questions is to add the suffix `_*_mc` after the original variable name; `*` represents the `*` option. For example: b1004_1_mc indicates whether the first option "food crop" of b1004 is ticked; 0 means not selected, 1 means selected.

Table 1 Processing Instruction of Acyclic Multiple-Choice Questions

varname_*_mc	Variable Information
0	The * option has not been selected
1	The * option has been selected

(2) Cyclic multiple-choice questions

The naming rule for cyclic multiple-choice questions is to add the suffix `_*_#_mc` after the original variable name; `*` represents the `*` option, while `#` represents `#` cycle. For example, b3002a_2_1_mc indicates whether the second option "Estimated that loan application will not be approved" is ticked in the first cycle; 0 means not selected, 1 means selected.

Table 2 Processing Instruction of Cyclic Multiple-Choice Questions

varname_*_#_mc	Variable Information
0	The * option has not been selected in the # cycle.
1	The * option has been selected in the # cycle.

6. Interpolation variable

In order to solve the problem of data missing, the question that the respondent did not answer the specific value and answered the numerical range (that is, the variable name suffix containing the

“it”) was interpolated. For the interpolated variable, the naming rule is the original variable name plus the suffix `_imp`.

The specific truncation variables for the household financial investigation data of 2011 and 2013 are detailed in Schedule 1-1 and Schedule 2-1.

7. Truncation

In accordance with the provisions of the statistical law, in order to protect the privacy of respondents, we have truncated the extreme values of income and assets. Replace the sample whose income or assets exceed the specific value with the specified value, and at the same time give a truncated dummy variable. The naming rule is to prefix the variable name with `censor_`, i.e. `censor_varname`. For example, we replaced the annual income (`hh_income`) of the 2011 China Household Finance Survey data with more than 3 million in 3 million, and also gave a dummy variable `censor_hh_income`, 0 means no truncation; 1 means truncation.

Table 3 Instruction to Truncation

<code>censor_varname</code>	Variable Information
0	Unprocessed
1	Processed

The specific truncation variables for the China Household Finance Survey data of 2011 and 2013 are detailed in Schedule 1-2 and Schedule 2-2.

8. Other special variables

(1) Instruction to individual variable in 2011

To facilitate research, we generated two new variables on income in the data of China Household Finance Survey of 2011, namely `labor_inc` (labor income) and `hh_income` (total income of individual).

1) `labor_inc`

`Labor_inc` refers to the individual after-tax wage and salary income, which is the labor income (including the sum of the most important work mentioned in the questionnaire and the after-tax wage and salary income of the second occupation).

2) hh_income

The hh_income refers to the total disposable income for the household in 2010, including the total after-tax cash income and in-kind income of all family members throughout the year. It is based on the household incomes in hh data plus the personal after-tax salary income (labor_inc) in ind data. It consists of five parts: wage and salary income, agricultural production income, industrial and commercial income, investment income and transfer income.

1. **Salary income:** after-tax cash income and in-kind income obtained from being employed by others. Including: after-tax wage income, after-tax bonus income, and after-tax subsidy income from the employer.
2. **Net income from agricultural production:** refers to the total net income after deducting the cost of agricultural production obtained from agricultural production, including the market value of agricultural products used for their own consumption.
3. **Net income from industry and commerce:** refers to the net income of industrial and commercial operations belonging to the household and obtained from the industrial and commercial production and operation projects after deducting the production and operation costs and taxes for households engaged in or participating in industrial and commercial production and operation.
4. **Investment income:** including rental income earned by rented houses, cars, etc., interest income from bank deposits, and net income from investment in financial products (including net incomes from investment in stocks, bonds, funds, futures, gold and other financial products, dividend income from investment insurance).
5. **Transfer income:** including various subsidized incomes obtained from the government, compensation for land acquisition and house demolition received by the household, retirement wages, housing accumulation fund withdrawn or used, insurance incomes (including unemployment insurance benefits compensated, commercial life insurance benefits, commercial health insurance compensation, commercial pension insurance benefits, other insurance compensations, etc.) and incomes from relationship with others.

Explanation: The hh_income income is negative because the household engaged in agricultural production and net income from industrial and commercial operations in 2010 was negative. A negative number in hh_income indicates that the household income is in a loss state. In the

household income, some households may be unable to make ends meet. For example, if the business is at a loss, the income may be negative.

(2) Instruction to individual variable in 2013

1) asset

The household's total assets include non-financial assets and financial assets. Non-financial assets include agricultural operating assets, industrial and commercial operating assets, land assets, real estate, vehicle assets, and other non-financial assets. Financial assets include social security account balances, cash, deposits, stocks, funds, bonds, derivatives, wealth management, foreign currency assets, gold, other financial assets, and lending.

2) total_income

Total income of a household includes wage income, agricultural operating income, industrial and commercial operating income, transfer income, and investment income.

3) Description of h2001 and h2002 variables

[h2001] Last year, did your household receive the following subsidies/allowance from the government, where the government subsidies do not include the agricultural production and operation subsidies mentioned above? (You may choose more than one option)

- | | |
|---|--|
| 1. No | 6. Relief payment, disaster relief payment |
| 2. Assistance payment for poor households | 7. Food allowance |
| 3. Bounty for one child household | 8. Subsidies for converting cultivated land into forests |
| 4. Assistance payment for households enjoying the five guarantees | 9. Others (please specify it)_____ |
| 5. Pension | |

[h2002] Last year, how much [CAPI autoloading H2001 items] has been received? (Unit: RMB)

[h2002 01] Assistance payment for poor households_____

[h2002 02] Bounty for one child household_____

[h2002 03] Assistance payment for households enjoying the five guarantees_____

[h2002 04] Pension_____

[h2002 05] Relief payment, disaster relief payment_____

[h2002 06] Food allowance_____

[h2002 07] Subsidies for converting cultivated land into forests_____

[h2002 08] Others (please specify it)_____

Please note that: the variable h2002_01 in the data corresponds to the second option in [h2001]: the assistance payment for poor households, h2002_02 corresponds to the third option in [h2001]: the bounty for one child household, and so forth.

(3) Instruction to questionnaire batches in 2013

The questionnaires for the first and second batches in 2013 were slightly deleted, and the variables for deletion were:

H3020, H3023, H3024, H3027, H3036, H3039, H3041, H3042

The first batch of investigation of the above variables has been completed with about 16,000 visits.

III. Schedules

1. Variable list in 2011

(1) Interpolation variable list in 2011

Schedule 1-1 Interpolation Variable List in 2011

Names of Variables				
c3016_#_imp	d5109_imp	d4103_#_imp	k1101_imp	a3036_imp
c2013_#_imp	d6106a_imp	d8104_#_imp	c7009_#_imp	f1010_imp
c2016_#_imp	d6110_imp	c7052a_imp	b1012_imp	f1031_imp
c2055_#_imp	d6115_imp	c7057_imp	b2013_imp	f2006_imp
d1105_imp	d6116_imp	g1018_imp	b2014_imp	f4008_imp
d2104_imp	d7106a_imp	c7050_#_imp	b2015_imp	f4011_imp
d3103_imp	d7110_imp	c7019_#_imp	b3005_imp	f1005_imp
d3109_imp	d7112_imp	g1014_imp	b1005_imp	f1022_imp
d3116_imp	d8106_imp	g1015_imp	b3033_imp	f1025_imp
d3117_imp	d9103_imp	h2004_imp	a3020_imp	f4005_imp

d5107_imp d9105_imp c8007_imp a3022_imp g1017_imp

(2) Truncation variable list in 2011

Schedule 1-2 Truncation Variable List in 2011

Names of Variables	Truncation Standard	Sample Amount
household data		
hh_income	3000000	7
hh_income	-500000	1
b2003b	2000000	15
b2003c	2000000	4
b2015_imp	2000000	8
b1005_imp	2000000	1
b1013a	2000000	4
c4007	200	2
c4008b	200	21
c6007	200	3
b1014	5000000	1
b2003a	5000000	15
b2013_imp	5000000	33
b2013	5000000	26
b2015	5000000	1
c5010_1	5000000	2
c5010_2	5000000	2
c8005_2	5000000	1
c8005_4	5000000	2
k2102a	5000000	1
c2016_2_imp	500	6
c2016_3_imp	500	2
c2016_1_imp	500	34
c2016_1	500	32
c2016_2	500	5
c2016_3	500	2
individual data		
labor_inc	2000000	1
f1031	2000000	1
f1031_imp	2000000	1

2. Variable list in 2013

(1) Interpolation variable list in 2013

Schedule 2-1 Interpolation Variable List in 2013

Names of Variables				
b1004a_imp	c2032_#_imp	d1111_imp	d7106a_imp	k1101_imp
b1005_imp	c2041_#_imp	d2104_imp	d7110_imp	k2102a_imp
b1005b_imp	c2055_#_imp	d2106_imp	d7112_imp	k2102b_imp
b1012_imp	c3002_#_imp	d2117_imp	d8104_#_imp	k2109_imp
b1013a_imp	c3002a_#_imp	d3103_imp	d8104b_imp	k2113_imp
b1014_imp	c3003_#_imp	d3109_imp	d8106_imp	a3020_imp
b2003b_imp	c3016_#_imp	d3110_imp	d9102_imp	a3022_imp
b2003c_imp	c3017c_imp	d3116_imp	d9103_imp	a3023_imp
b2013_imp	c3021_#_imp	d3117_imp	d9105_imp	a3024_imp
b2014_imp	c3022_#_imp	d4103_#_imp	e1007b_imp	a3036_imp
b2015_imp	c7000a_#_imp	d4104_#_imp	e1021_imp	a3037_imp
b3003a_#_imp	c7000c_#_imp	d4108_#_imp	e1022_imp	f1005_imp
b3005_#_imp	c7000e_#_imp	d4111_imp	e1027_imp	f1010_imp
b3008_#_imp	c7009_#_imp	d5107_imp	e1034_imp	f1018a_imp
b3031_#_imp	c7015a_#_imp	d5108_imp	e2020_imp	f1022_imp
b3031a_#_imp	c7019_#_imp	d5109_imp	e3004b_imp	f1025_imp
b3044_#_imp	c7047b_#_imp	d6106a_imp	e3004c_imp	f1031_imp
b3051_#_imp	c7047d_#_imp	d6106b_imp	e3007a_imp	f1031a_imp
c2000a_#_imp	c7048b_#_imp	d6110_imp	e3014_imp	f2006_imp
c2000c_#_imp	c7048k_#_imp	d6111_imp	g1014_imp	f4005_imp
c2000f_#_imp	c7051b_imp	d6115_imp	g1015_imp	f4008_imp
c2013_#_imp	c7052a_imp	d6115c_imp	g1017_imp	f4011_imp
c2013a_#_imp	c8007_imp	d6116_imp	g1018_imp	
c2016_#_imp	d1105_imp	d7105_imp	h2004_imp	

(2) Truncation variable list in 2013

Schedule 2-2 Truncation Variable List in 2013

Names of Variables	Truncation Standard	Sample Amount
asset	2000w	38
b1004a	300w	7
b1004a_imp	300w	7
b1005_imp	300w	6
b1005b	200w	2

b1005b_imp	200w	2
b1012	200w	7
b1012_imp	200w	8
b1013a	200w	2
b1013a_imp	200w	2
b1014_imp	500w	1
b2003a	300w	46
b2003b_imp	300w	27
b2003c_imp	300w	5
b2013_imp	800w	66
b2014_imp	800w	38
b2015_imp	300w	13
b3003a_1	300w	9
b3003a_1_imp	300w	9
b3003a_2	500w	5
b3003a_2_imp	500w	5
b3005_1	300w	1
b3005_1_imp	300w	1
b3005_2	300w	3
b3005_2_imp	300w	3
b3008_1	200w	1
b3008_1_imp	200w	1
b3031_1	500w	8
b3031_1_imp	500w	10
b3031_2	200w	3
b3031_2_imp	200w	3
b3031a_1	200w	2
b3031a_1_imp	200w	2
b3031a_2	200w	1
b3031a_2_imp	200w	1
b3044_1	500w	21
b3044_1_imp	500w	21
b3044_2	300w	7
b3044_2_imp	300w	7
b3051_1	300w	7
b3051_1_imp	300w	7
b3051_2	300w	1
b3051_2_imp	300w	1
c2000a_1	500w	15
c2000a_1_imp	500w	27
c2000a_2	300w	6
c2000a_2_imp	300w	7
c2000a_3	300w	1
c2000a_3_imp	300w	1
c2000f_1	300w	1
c2000f_1_imp	300w	1
c2000f_2	200w	2
c2000f_2_imp	200w	2
c2013_1	800w	88
c2013_1_imp	800w	87
c2013_2	500w	27

c2013_2_imp	500w	28
c2013_3	300w	3
c2013_3_imp	300w	3
c2013a_1	200w	4
c2013a_1_imp	200w	4
c2016_1_imp	800w	20
c2016_2_imp	500w	24
c2016_3	300w	16
c2016_3_imp	300w	17
c2032_1	300w	1
c2032_1_imp	300w	1
c2032_2	200w	1
c2032_2_imp	200w	1
c2041_1	200w	5
c2041_1_imp	200w	5
c2041_2	300w	1
c2041_2_imp	300w	1
c2055_1	200w	2
c2055_1_imp	200w	2
c2055_2	300w	1
c2055_2_imp	300w	1
c3002_1_imp	200w	5
c3003_1	800w	27
c3003_1_imp	800w	27
c3003_2	200w	2
c3003_2_imp	200w	2
c3016_1	200w	6
c3016_1_imp	200w	6
c3017c	300w	23
c3017c_imp	300w	34
c3021_1	200w	3
c3021_1_imp	200w	3
c3021_3	200w	1
c3021_3_imp	200w	1
c3022_1	300w	11
c3022_1_imp	300w	12
c3022_2	300w	2
c3022_2_imp	300w	2
c4007a	200w	1
c5010_1	300w	44
c5010_2	300w	6
c5010_3	300w	7
c5010_4	300w	1
c7000a_1	300w	1
c7000a_1_imp	300w	1
c7000e_1	200w	2
c7000e_1_imp	200w	2
c7009_2	300w	1
c7009_2_imp	300w	1
c7015a_2	200w	1
c7015a_2_imp	200w	1

c7051b	200w	1
c7051b_imp	200w	1
c7052a	200w	1
c7052a_imp	200w	1
c8005_1	300w	1
c8005_2	300w	2
c8005_6	300w	1
c8007	200w	1
c8007_imp	200w	1
d1105	300w	3
d1105_imp	300w	3
d1111	300w	2
d1111_imp	300w	1
d2104	300w	1
d2104_imp	300w	1
d2117	200w	1
d2117_imp	200w	1
d3103	200w	1
d3103_imp	200w	1
d3109	200w	3
d3109_imp	200w	3
d3110	200w	6
d3110_imp	200w	6
d3116	200w	1
d3116_imp	200w	1
d4103_1	200w	1
d4103_1_imp	200w	1
d4104_1	200w	1
d4104_1_imp	200w	1
d4108_1	200w	1
d4108_1_imp	200w	1
d5108	300w	1
d5108_imp	300w	1
d7105	300w	2
d7105_imp	300w	1
d7106a	300w	1
d7106a_imp	300w	1
d7110	200w	1
d7110_imp	200w	1
d9103	300w	2
d9103_imp	300w	1
e3004b	200w	1
e3004b_imp	200w	2
e3007a	200w	1
e3007a_imp	200w	1
e3014	200w	1
e3014_imp	200w	1
h2004_imp	200w	1
k1101	200w	1
k1101_imp	200w	1
k2102a	300w	7

k2102a_imp	300w	5
k2102c	200w	6
k2102c_imp	200w	6
k2109	300w	1
k2109_imp	300w	1
total_income	500w	8
total_income	-100w	7
total_income_imp	500w	8
total_income_imp	-100w	7

3. List of other special variables

Schedule 3-1 Special Variable List

Names of Variables	Variable Value	Meaning of Variable Value	Supplementary Instruction
hhead	0	Not the householder	In the ind data set
	1	Householder	
rural	0	Urban	In the master data set
	1	Rural	
region	1	East China	Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, Hainan
	2	Central China	Shanxi, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, Hunan
	3	West China	Inner Mongolia, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, Qinghai, Ningxia